Assignment 2 Report

MNIST Classification using PCA, FDA, LDA and QDA

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1. Implementation

Step 1 — Data Preparation

I loaded the MNIST dataset and kept only digits 0, 1 and 2.
From each class, I randomly selected 100 samples for training and 100 for testing, so both train and test sets have 300 samples each.

Each 28×28 image was flattened into a 784-dimensional vector and normalised to the range [0, 1].

Step 2 — PCA (Implemented from Scratch)

I implemented PCA manually using NumPy.

Steps followed:

1. Formed the data matrix X of shape (784, 300).
2. Computed the mean `mu` and centred the data:

   Xc = X - mu

3. Computed covariance matrix:

   S = (Xc @ Xc.T) / (n - 1)

4. Used `np.linalg.eigh` to compute eigenvalues and eigenvectors.
5. Sorted eigenvalues in descending order.
6. Computed cumulative variance ratio:

variance_ratio = eigvals / sum(eigvals)

Number of Components Used

- 19 components → retain 75% variance
- 51 components → retain 90% variance
- 2 components → only first two PCs (for visualisation)

Projection was done using:

Y_train = Up.T @ Xc

For the test set, I subtracted the **training mean only**, to avoid data leakage.

Reconstruction

To check how much information PCA retained, I reconstructed the images:

X_hat = Up @ Y_train + mu

I computed MSE for the first 5 samples:

Sample	MSE
1	0.01694
2	0.02019
3	0.01506
4	0.01625
5	0.01481
	Mean : .0166

The errors are small, which shows that even 19 components capture most of the important structure in the images.

Step 3 — Fisher's Discriminant Analysis (FDA)

FDA was applied on the PCA (75%) projected data (19D), not directly on 784D.
If we apply FDA directly in 784D with only 300 samples, SW becomes singular.

Scatter Matrices

Between-class scatter:

SB = sum Nc * (mu_c - mu)(mu_c - mu).T

Within-class scatter:

SW = sum over classes of sum (x - mu_c)(x - mu_c).T

Then I solved:

SW_inv @ SB @ w = lambda * w

Since there are 3 classes:

rank(SB) ≤ C - 1 = 2

So only 2 meaningful discriminant directions exist.

Final FDA projection gives 2D data:

Z_train shape = (300, 2)

Step 4 — Classification (LDA & QDA)

Both LDA and QDA were implemented manually.

LDA

- Uses shared covariance matrix.
- Linear decision boundary.

QDA

- Uses separate covariance per class.
- Quadratic decision boundary.

- More flexible than LDA.

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2. Accuracy Results

(Please note I have assumed we only need to perform these 5 cases)

Case	Pipeline	Train Accuracy	Test Accuracy
1	PCA75 → FDA → LDA	97.00%	94.33%
2	PCA75 → FDA → QDA	99.00%	96.33%
3	PCA75 → LDA	97.00%	94.33%
4	PCA90 → LDA	99.67%	96.00%
5	PCA2 → LDA	92.67%	92.33%
6	FDA (raw) → LDA	100.00%	75.67%
7	FDA (raw) → QDA	100.00%	48.33%

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3. Observations

Effect of PCA Components

- Using only 2 PCs reduces accuracy (~92% test).
- 19 PCs (75% variance) already performs well (~94% test).
- 51 PCs (90% variance) improve performance further (~96% test).

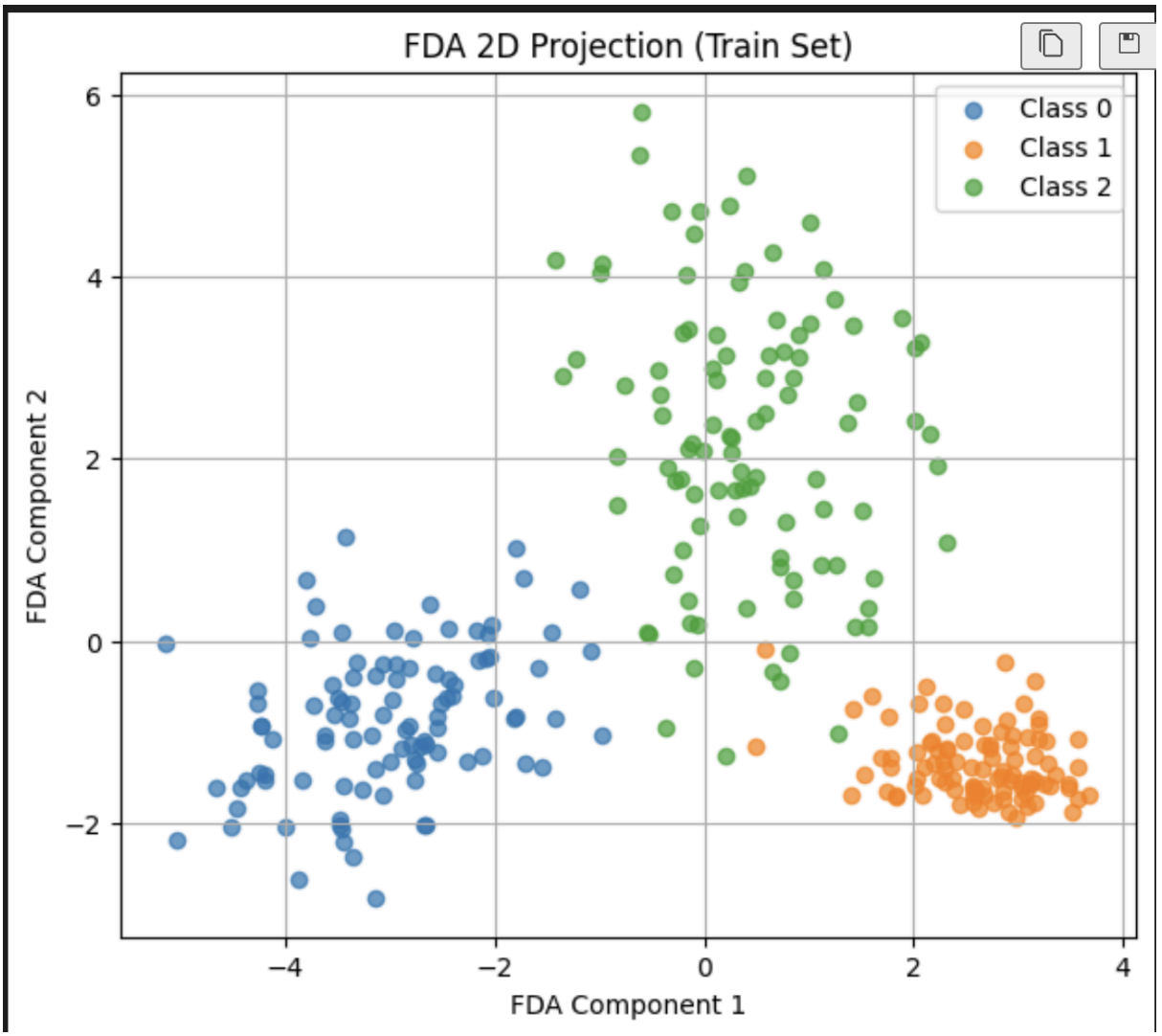
So, increasing retained variance generally improves LDA performance.

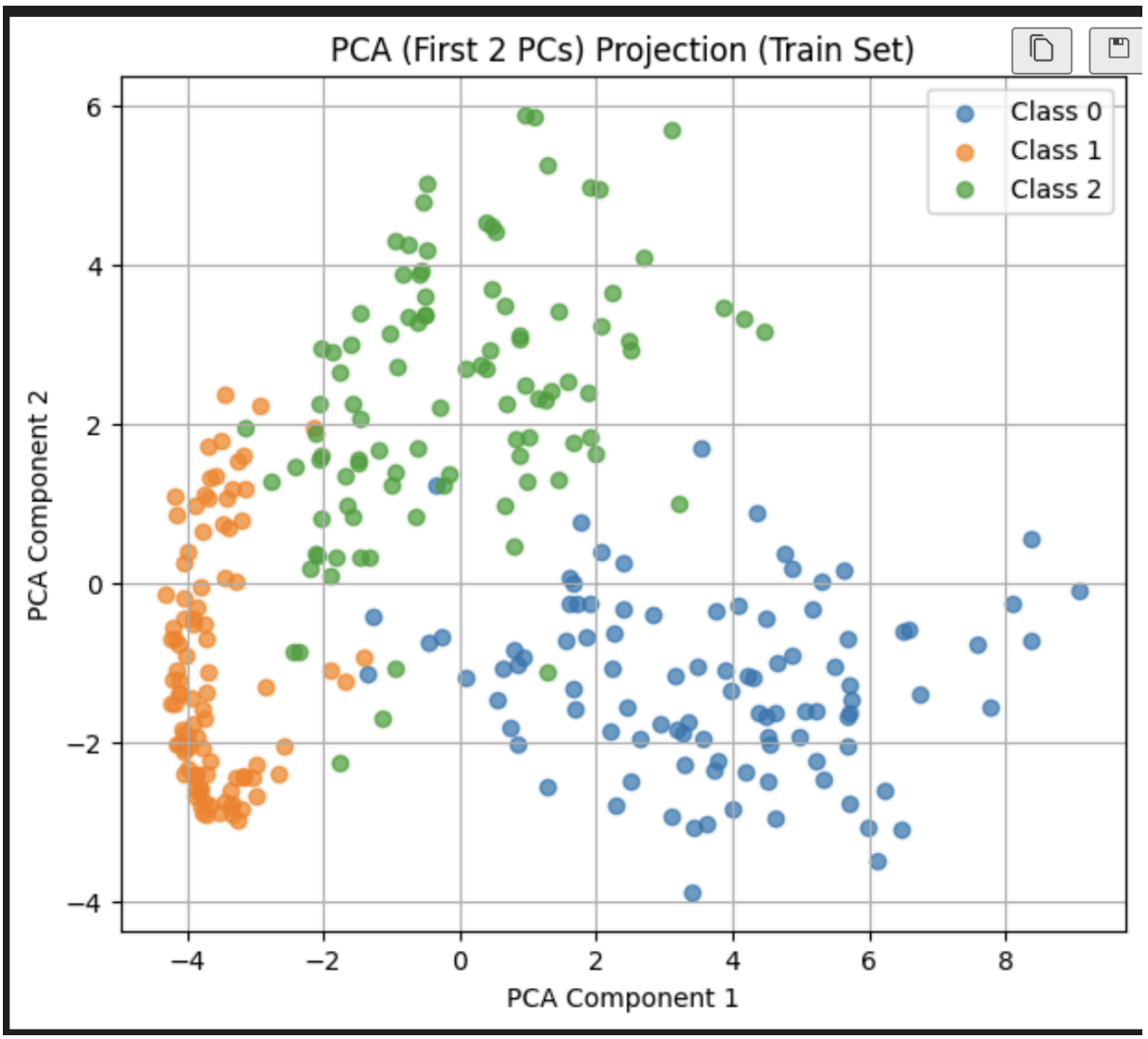
FDA vs PCA

When plotting 2D projections:

- FDA shows much clearer separation between classes.
- PCA (2D) shows noticeable overlap.

Plots :





This makes sense because:

- PCA is unsupervised (maximises variance).
- FDA is supervised (maximises class separability).

Even though FDA improves visual separation, PCA75 already retains strong class information, which is why PCA75 → LDA and PCA75 → FDA → LDA give similar accuracy.

Also, note that performing FDA on raw data gives 100% accuracy on the train set, but a very low accuracy on the test set. This is the typical case of overfitting, as it trains on 784 parameters, leading to a very high train accuracy. Applying PCA first, reduces this.

LDA vs QDA

QDA consistently performs slightly better than LDA.

Reason:

- QDA allows class-specific covariance.
- In 2D FDA space, QDA does not overfit much.

Generalisation

Train and test accuracies are close in all cases, so there is no significant overfitting.

Final Conclusion

In this assignment:

- PCA was used for dimensionality reduction.
- FDA was used for supervised projection.
- LDA and QDA were used for classification.
- Performing FDA on raw data leads to overfitting.

Main takeaways:

- Increasing PCA components improves accuracy.
- QDA performs slightly better than LDA.
- FDA produces better visual class separation than PCA.
- Even strong dimensionality reduction still retains meaningful classification performance.